

Schur-functor representations of $O(n)$

Plethystic pushforward, polynomial-Gaussian traces, and matrix verification

Proof and verification note

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Outcome. The orthogonal formula is correct as a coefficientwise stable plethystic pushforward. The finite trace identity is exact before taking a limit. Five representative (n, λ) cases give 40 direct comparisons with maximum error 5.33×10^{-15} . Haar $O(9)$ tests recover the invariant quadratic form in $\text{Sym}^2 W$, its absence from $\Lambda^2 W$, and the odd-degree parity obstruction.

1 Stable sigma-MGF and its proof

Let $W_n = \mathbb{C}^n$ with the defining action of the compact group $O(n)$, fix $\lambda \vdash d$, and put $V_{\lambda,n} = S_\lambda(W_n)$. For a partition τ ,

$$[m_\tau] \mathcal{S}_{\lambda,n}^O = \dim \left(\bigotimes_i \text{Sym}^{\tau_i} S_\lambda(W_n) \right)^{O(n)} = \mu_{O(n)}(h_\tau[s_\lambda]). \quad (1)$$

The second equality is simply associativity of plethysm: $h_\tau[s_\lambda[X]] = (h_\tau[s_\lambda])[X]$. The defining-representation stable sigma-MGF is $\text{Exp}_\sigma(h_2)$. Applying its coefficient functional to the fixed symmetric function $h_\tau[s_\lambda]$ proves

$$[m_\tau] \mathcal{S}_{\lambda,\infty}^O = \langle \text{Exp}_\sigma(h_2), h_\tau[s_\lambda] \rangle. \quad (2)$$

No new limiting invariant theory is needed after the defining case. A conservative sufficient stable range is $n \geq d|\tau|$: the relevant expression has tensor degree $d|\tau|$, where Brauer pairings are independent in that range.

Since $-I \in O(n)$ acts on $S_\lambda(W_n)$ by $(-1)^d$, it acts on the space in (1) by $(-1)^{d|\tau|}$. Therefore

$$[m_\tau] \mathcal{S}_{\lambda,n}^O = 0 \quad \text{if } d|\tau| \text{ is odd.} \quad (3)$$

This parity conclusion is exact at finite n .

2 Exact traces and the limiting law

For every $g \in O(n)$ and $r \geq 1$, functoriality gives $S_\lambda(g)^r = S_\lambda(g^r)$, whence

$$\text{Tr}(S_\lambda(g)^r) = s_\lambda[X(g^r)] = \sum_{\alpha \vdash d} \frac{\chi^\lambda(\alpha)}{z_\alpha} \prod_{a \in \alpha} \text{Tr}(g^{ra}). \quad (4)$$

This formula is exact for every allowed n ; stability is needed only when replacing traces by Gaussian variables.

The joint moment convergence of finitely many orthogonal trace powers is an external classical-group trace theorem (Diaconis–Evans, with the normalization below). Let independent $A_k \sim N(0, 1)$ and set

$$Z_k^O = \sqrt{k} A_k + \mathbf{1}_{2|k}. \quad (5)$$

Joint moment convergence of fixed orthogonal trace powers and polynomial substitution in (4) give

$$\mathrm{Tr}(S_\lambda(g)^r) \longrightarrow Q_{\lambda,r}^O = \sum_{\alpha \vdash d} \frac{\chi^\lambda(\alpha)}{z_\alpha} \prod_{a \in \alpha} Z_{ra}^O. \quad (6)$$

For $\lambda = (2)$, $Q_{(2),1}^O = \frac{1}{2}((Z_1^O)^2 + Z_2^O)$ has mean 1, recording the invariant quadratic form. For $\lambda = (1, 1)$, the minus sign makes the mean zero.

3 Matrix construction and formula cross-check

The verification draws seeded Haar orthogonal matrices by real Gaussian QR with sign correction. A standard Young symmetrizer c_λ is built explicitly on $W_n^{\otimes d}$. If B is an orthonormal basis of $\mathrm{im} c_\lambda$, the direct representation matrix is

$$D_\lambda(g) = B^* g^{\otimes d} B. \quad (7)$$

The formula cross-check never uses this matrix: it computes the defining traces, uses Newton’s recurrence $j h_j = \sum_{k=1}^j p_k h_{j-k}$, and evaluates the Jacobi–Trudi determinant $s_\lambda = \det(h_{\lambda_i - i + j})$. Tests compare $\mathrm{Tr}(D_\lambda(g)^r)$ for $r = 1, 2, 3$. They also compare

$$\prod_i h_{\tau_i}[D_\lambda(g)] \quad \text{against} \quad \prod_i h_{\tau_i}[s_\lambda][X(g)] \quad (8)$$

for $\tau = (), (1), (2), (1, 1), (3)$, so the actual sigma-MGF integrands are covered.

n	tested λ	checks	purpose	status
3	$(1), (2), (3)$	trace and sigma integrands	defining/symmetric powers	PASS
4	$(1, 1), (2, 1)$	trace and sigma integrands	alternating/mixed symmetry	PASS
40 comparisons; maximum absolute error				5.33×10^{-15}

The seeded Haar diagnostic uses 1,200 matrices in $O(9)$. It gives means 0.9979 for $\chi_{(2)}$ (target 1), -0.0289 for $\chi_{(1,1)}$ (target 0), and 0.0116 for $\chi_{(3)}$ (target 0). Each lies inside 4.5 estimated standard errors. Group residuals are at most 3.24×10^{-15} .

4 Reproducibility and interpretation

Run the module below from the repository root, or open `orthogonal/notebook.py` with marimo. Exact rows are the proof-driven regression test. Haar rows are seeded numerical diagnostics and are reported with standard errors; they are not presented as proofs of convergence.

```
python -m for_this_guy.schur_functor_classical_groups.orthogonal.verification
```

References

- S. Howe, *Random matrix statistics and zeroes of L -functions via probability in λ -rings*.
G. Lehrer and R. Zhang, *The Brauer Category and Invariant Theory*.
P. Diaconis and S. N. Evans, *Linear functionals of eigenvalues of random matrices*, Trans. Amer. Math. Soc. 353 (2001), 2615–2633.